

Problem 1

In this problem, we use the OnlineNews dataset available on Canvas. The dataset is about predicting the popularity of websites from 58 features. Discard the first two variables from the data. From the remaining variables use all but the last variable as predictors and the log of the last variable.

Obtain all results using 5-fold cross-validation, which is computed as follows:

1. Generate a random permutation of the data. Use this randomized permutation to split the data into 5 disjoint subsets of almost equal size.
2. For each fold $i \in \{1, \dots, 5\}$, train the model on all data except subset i and test it on subset i , obtaining test error ϵ_i .
3. Obtain the final test error as the average of the 5 test errors ϵ_i obtained above.

Report results for the following models:

- Null model. Report the average train and test MSE of the null model that always predicts training $\bar{y}$ (average training y)

We do this in Python and obtain a train MSE of 0.8657 and a test MSE of 0.8658.

- OLS regression computed analytically by solving the following normal equations:

$$\left(\frac{1}{N}X^T X + \lambda I_p\right)\beta = \frac{1}{N}X^T Y$$

where $\lambda \in \{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ and N is the number of observations in X . Report in a table the average training and test R^2 and MSE, as well as their standard deviations obtained from the 5 folds. On the same graph, plot the average training and test MSE vs λ as two separate curves. Also plot the average value of the logarithm of the determinant of $\frac{1}{N}X^T X + \lambda I_p$ vs λ .

We do this in Python and obtain table 1, figure 1, and figure 2

Lambda	Train R2	Test R2	Train MSE	Test MSE
0.000000	0.127 (0.001)	-0.001 (0.274)	0.756 (0.004)	0.867 (0.244)
0.000010	0.127 (0.001)	-0.063 (0.413)	0.756 (0.004)	0.922 (0.366)
0.000100	0.127 (0.001)	-0.202 (0.725)	0.756 (0.004)	1.044 (0.638)
0.001000	0.127 (0.001)	-0.035 (0.351)	0.756 (0.004)	0.897 (0.311)
0.010000	0.126 (0.001)	0.108 (0.029)	0.757 (0.004)	0.772 (0.033)
0.100000	0.118 (0.001)	0.114 (0.006)	0.763 (0.004)	0.767 (0.017)

Table 1: Mean (SD) of MSE and R^2 5 fold CV. Ridge-like Regression.

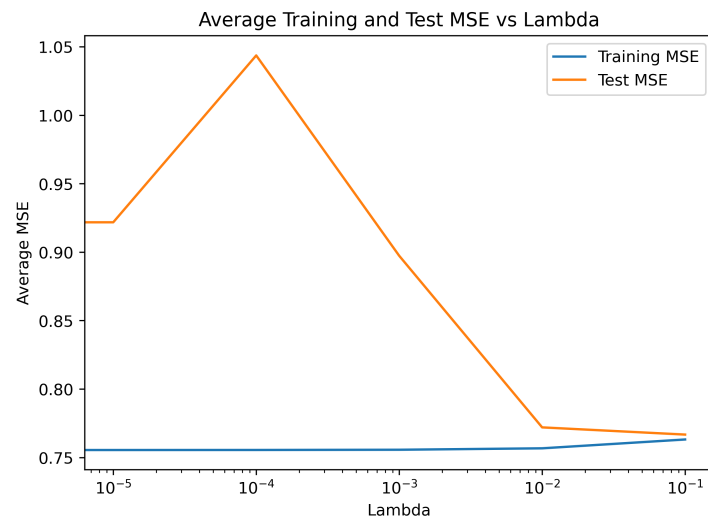


Figure 1: Average training and testing MSE vs Lambda.

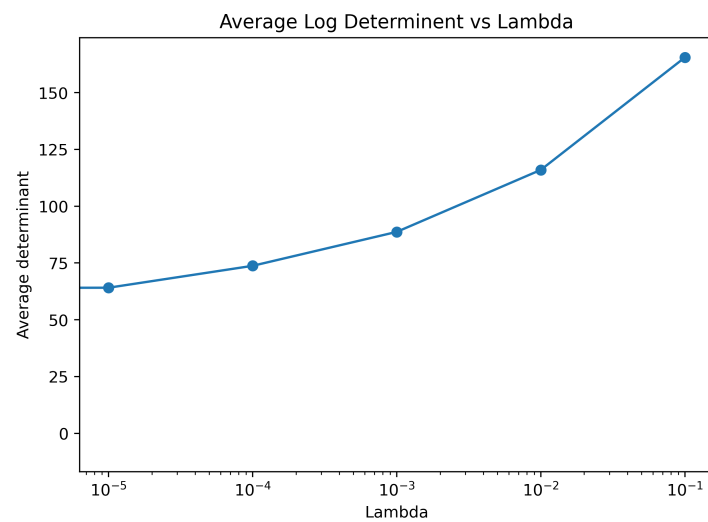


Figure 2: Average log determinant vs Lambda.

- OLS regression with backward elimination, for example using SequentialFeatureSelector. For each number of features $k \in \{5, 10, 15, \dots, 55\}$ report the average training and test R^2 and MSE in a table. On the same graph, plot the average training and test R^2 vs k as two separate cruves.

Doing this, we get table 2 and figure 3.

k	train R^2	test r2
5	0.0823	0.0806
10	0.1059	0.1032
15	0.1136	0.1112
20	0.1174	0.1137
25	0.1200	0.1156
30	0.1212	0.1164
35	0.1221	-1.6069
40	0.1226	-1.6933
45	0.1230	0.0895
50	0.1233	0.0562
55	0.1250	0.0877

Table 2: Average Training and Test R^2 for Backward Feature Selection

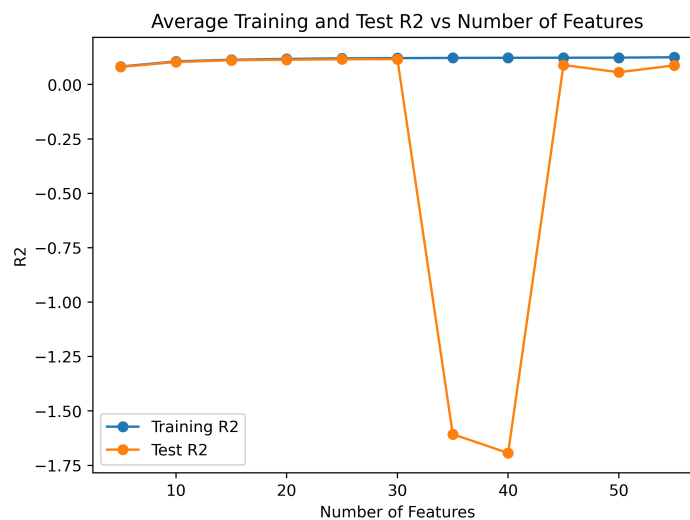


Figure 3: Average training and test R^2 vs k .

The negative test R^2 values are odd, but not impossible given what R^2 is measuring. The definition of R^2 is:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

For R^2 to be negative, all we need is for $\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$ to be greater than 1. In other words, $\sum_{i=1}^n (y_i - \hat{y}_i)^2 > \sum_{i=1}^n (y_i - \bar{y})^2$, assuming $\sum_{i=1}^n (y_i - \bar{y})^2 > 0$, which should be appropriate if all y_i are not identical (aka if $\bar{y} = y_i \quad \forall i$).

To demonstrate this more concretely, what if we let all $\hat{y}$ be a constant c ? In that case:

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - c)^2$$

and if we complete the square, we get:

$$\sum_{i=1}^n (y_i - c)^2 = \sum_{i=1}^n (y_i - \bar{y})^2 + n(\bar{y} - c)^2$$

plugging this into the formula for R^2 , we get:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \bar{y})^2 + n(\bar{y} - c)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{-n(\bar{y} - c)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

This is 0 if $c = \hat{y}$, negative for any other value of c , and there is no clear bound on how negative it can go. In other words, a negative R^2 value is possible if the predictions are far from $\bar{y}$.

- Random forest regression with 100 trees and maximum depths 1, 2, ..., 10. Report the average training, OOB, and test R^2 and MSE and their standard deviations in each case. Also, plot the average training, OOB, and test R^2 vs the maximum depth as three separate curves.

In Python, we can do this and obtain table 3 and figure 4:

Depth	Train R2	Test R2	Train MSE	Test MSE	OOB R2	OOB MSE
1	0.055 (0.001)	0.054 (0.004)	0.818 (0.005)	0.819 (0.018)	0.054 (0.001)	0.819 (0.005)
2	0.082 (0.001)	0.079 (0.006)	0.795 (0.004)	0.797 (0.019)	0.079 (0.001)	0.797 (0.004)
3	0.104 (0.001)	0.099 (0.006)	0.775 (0.004)	0.78 (0.019)	0.099 (0.001)	0.78 (0.004)
4	0.124 (0.001)	0.114 (0.006)	0.758 (0.004)	0.767 (0.018)	0.113 (0.001)	0.768 (0.004)
5	0.145 (0.001)	0.124 (0.006)	0.741 (0.004)	0.758 (0.018)	0.124 (0.001)	0.759 (0.004)
6	0.17 (0.001)	0.132 (0.005)	0.718 (0.004)	0.751 (0.018)	0.131 (0.001)	0.752 (0.004)
7	0.203 (0.001)	0.139 (0.005)	0.69 (0.004)	0.746 (0.017)	0.137 (0.001)	0.747 (0.004)
8	0.244 (0.002)	0.143 (0.004)	0.654 (0.004)	0.742 (0.017)	0.142 (0.001)	0.743 (0.004)
9	0.293 (0.003)	0.147 (0.004)	0.612 (0.004)	0.738 (0.017)	0.144 (0.001)	0.741 (0.004)
10	0.348 (0.004)	0.15 (0.004)	0.564 (0.005)	0.736 (0.017)	0.146 (0.001)	0.739 (0.004)

Table 3: Mean (SD) of MSE and R^2 for various tree depths for a random forest of 100 trees.

The average OOB R^2 almost exactly matches the test R^2



Figure 4: Average Training, OOB, and Test R^2 vs Maximum Depth.